

Group 2 - Logistic Regression Project Report

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Project Information

Project Title: Logistic Regression and Log-Loss Implementation
Course: CSCN8010 - Machine Learning Fundamentals
Instructor: Professor David
Date: February 2026

Group 2 - Team Members

Name	Student ID	Role	Contribution
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Ali Cihan Ozdemir	9091405	Lead Developer & Presenter	100% - Full project implementation
Lohith Reddy Danda	9054470	Co-Developer & Presenter	Documentation & Code Review
Roshan Bartaul	N/A	*(Did not contribute - No attendance)*	0%

Project Overview

What is This Project About?

This project covers the implementation and understanding of **Logistic Regression** as a fundamental statistical classification algorithm in machine learning. The main focus is on binary classification problems where the goal is to predict one of two possible outcomes (e.g., Pass/Fail, Yes/No).

Key Topics Covered:

- Statistical Classification** - Understanding classification problems in machine learning
- Logistic Regression vs Linear Regression** - Why logistic regression is needed for classification

- 3. **The Sigmoid Function** - The mathematical foundation: $\sigma(z) = 1 / (1 + e^{(-z)})$
- 4. **Log-Loss (Cross-Entropy)** - The loss function used to evaluate probabilistic predictions
- 5. **Model Evaluation** - Accuracy, Precision, Recall, F1-Score, and Confusion Matrix

Project Deliverables:

- * **Presentation Notebook** (`Group2_LogisticRegression_Presentation.ipynb`) - Used for Teams meeting presentation
- * **Assignment Notebook** (`Group2_LogisticRegression_Assignment.ipynb`) - Complete assignment with tasks

GitHub Repository

Group 2 Repository: https://github.com/alicih4n/LogisticRegressionClassifier_Workshop.git

Peer Review - Group 5

As part of the course requirements, this project was reviewed by Group 5.

Group 5 - Team Members

	Name		Student ID	
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	Preeja Anilal		8791796	
	Anthony Nosa Izevbokun		9016626	

Group 5 Repository

Group 5 Repository: https://github.com/AntUse/LogisticRegressionClassifier_Workshop_Group5.git

Key Learning Outcomes

1. Understanding the difference between linear and logistic regression
 2. Implementing the sigmoid function for probability estimation
 3. Building and training logistic regression models using scikit-learn
 4. Understanding and implementing the Log-Loss (Cross-Entropy) function
 5. Evaluating model performance using various metrics (Accuracy, Precision, Recall, F1-Score)
6. Visualizing decision boundaries and model results

Technical Details

Libraries Used:

- * NumPy - Numerical computing
- * Matplotlib - Visualization
- * Scikit-learn - Machine learning algorithms
- * Seaborn - Statistical graphics

Dataset Example:

The project uses a "Hours Studied vs. Pass/Fail" dataset to demonstrate logistic regression concepts in a simple and understandable way.

Conclusion

This project successfully demonstrates the fundamental concepts of logistic regression as a binary classification technique. The presentation covers the mathematical foundations, practical implementation, and evaluation methods essential for understanding modern machine learning classification algorithms.

Submitted for CSCN8010 - Machine Learning Fundamentals
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